

Student Sleep & Academic Performance

1. Project Goal & Research Questions

Sleep deprivation is increasingly common among university students, and its impact on concentration, motivation, and academic results is hard to ignore. While previous research on this dataset (e.g. Abdullah & Rahman's psychometric modeling paper) has used it for predictive ML models, we want to take a different angle: rather than predicting GPA, we want to give students and educators an interactive way to explore these patterns themselves, without needing any statistical background.

Concretely, the goal of this project is to build a single-page visualization dashboard that loads the real dataset and lets users filter by gender, year of study, and lifestyle habits to visually answer three research questions:

- **RQ1 — Sleep quality & academic difficulties:** Do students with poor sleep quality or insomnia symptoms report more concentration difficulties, higher fatigue, and lower academic performance?
- **RQ2 — Sleep duration & perceived performance:** Is there a visible relationship between the number of hours slept per night and self-reported academic performance level?
- **RQ3 — Lifestyle predictors of poor sleep:** How do screen time before bed, caffeine consumption, stress levels, and physical activity correlate with insomnia symptoms and sleep quality ratings?

2. Visualization Sketches & Design Rationale

We split the visualizations into two tiers. The **MVP (core)** charts directly answer the three research questions and are required for the final product. The **stretch goals** add richer exploration but can be dropped without breaking the project's coherence.

2.1 Core — VIZ 1: Sleep Quality × Academic Performance (Heatmap)

The heatmap is the most direct answer to RQ1. Rows = sleep quality levels (Poor to Excellent), columns = performance levels (Below Average to Excellent), and cell color encodes the count of students in each combination. A strong diagonal pattern, dark cells running from top-left (poor sleep, poor performance) to bottom-right (good sleep, good performance), would confirm the core hypothesis. Unlike a scatter plot, the heatmap works naturally with ordinal data and makes the overall correlation pattern immediately readable.

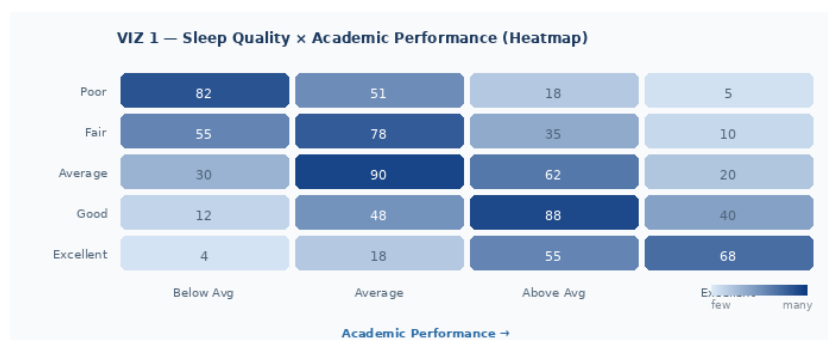


Figure 1 — Sketch: Sleep Quality × Academic Performance heatmap. Color intensity = number of students. Hover tooltip shows exact count.

Tool: D3.js v7 (scaleBand, scaleSequential, color interpolation d3.interpolateBlues) · **Interaction:** hover tooltip + row/column highlight

2.2 Core — VIZ 2: Sleep Duration × Academic Performance (Grouped Bar Chart)

A grouped bar chart shows how students in each sleep duration bucket (<5h, 5–6h, 6–7h, 7–8h, >8h) distribute across the four performance categories. This directly addresses RQ2. Compared to a stacked bar, the grouped layout preserves the ability to read individual bar heights — important here because we care about trends within each performance level across duration buckets, not just total proportions.

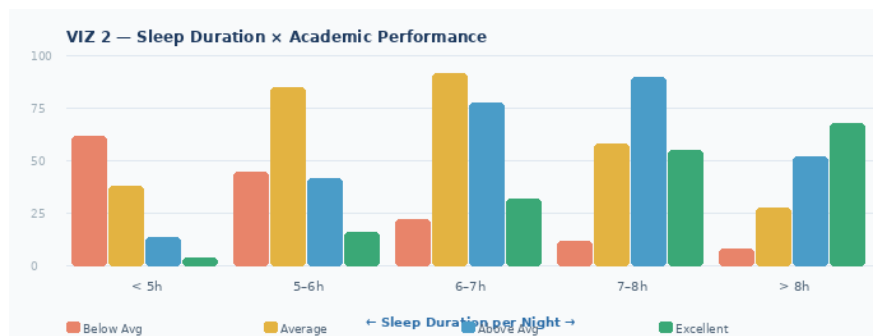


Figure 2 — Sketch: Grouped bar chart, sleep duration vs. academic performance distribution. Click a duration group to isolate it.

Tool: D3.js v7 (nested scaleBand, grouped bars, d3-transition) · **Interaction:** click to isolate a group, hover for count tooltip

2.3 Core — VIZ 3: Lifestyle Profile Parallel Coordinates

Parallel coordinates is the most expressive visualization for RQ3 because it shows multi-variable patterns simultaneously. Each **line represents one student**, traced across five axes: Stress Level, Screen Time, Caffeine, Sleep Quality, and Academic Perf. Lines are colored by sleep quality tier. The visual goal is to see whether the orange/red lines (poor sleepers) cluster at the top of the Stress and Screen Time axes — which would confirm those lifestyle factors as correlates of poor sleep.

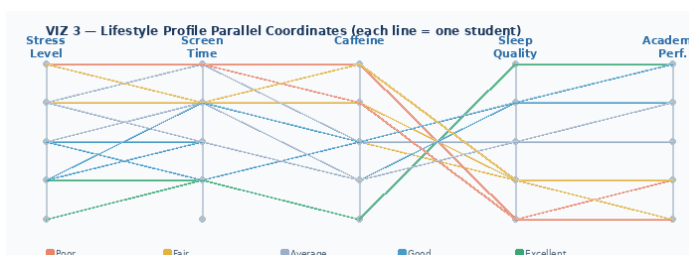


Figure 3 — Sketch: Parallel coordinates. Each line = one student. Color = sleep quality. Brushing one axis filters all lines.

Tool: D3.js v7 (scalePoint, multi-axis path generator, d3-brush per axis) · **Interaction:** hover to highlight line, brush any axis to filter

2.4 Stretch Goal 1 — Sankey Diagram (Duration → Quality → Performance)

The Sankey diagram adds a **flow perspective** that the heatmap and bar chart cannot provide: you can see the full journey of a student cohort across three variables simultaneously. Band widths are proportional to student counts, so dominant paths become visually obvious — e.g. a thick band from '6–7h' → 'Average' → 'Above Average' would suggest that moderate sleep is linked to above-average performance.

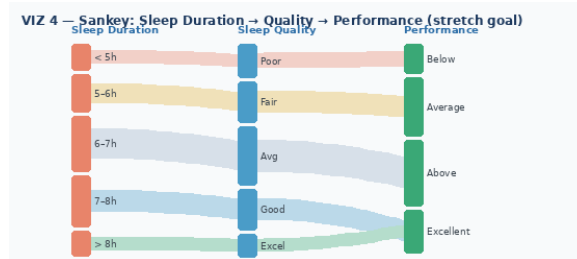


Figure 4 — Sketch: Sankey flow diagram (stretch goal). Band width $\propto$ number of students on that path.

Tool: d3-sankey plugin + D3.js v7

2.5 Stretch Goal 2 — Radar Chart Lifestyle Comparison

Two radar charts side by side let users select any two subgroups (e.g. Year 1 vs. Year 4, Male vs. Female, or Poor Sleepers vs. Good Sleepers) and compare their lifestyle profiles across five axes: Stress, Screen Time, Caffeine, Fatigue, and Physical Activity. The overlap between the two polygons immediately reveals similarities and differences.

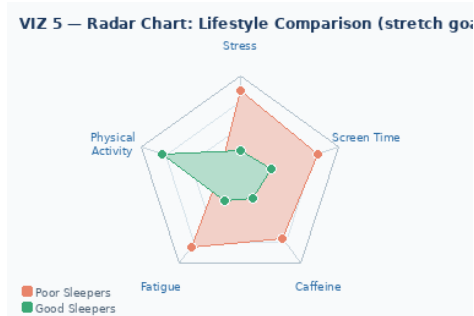


Figure 5 — Sketch: Radar chart comparing Poor Sleepers (orange) vs. Good Sleepers (green) across 5 lifestyle axes.

Tool: Chart.js (radar type) or D3.js custom polar layout

3. Tools & Relevant Lectures

Tool / Library	Role in this project	Used for	Lecture(s)
D3.js v7	Custom heatmap, grouped bar, parallel coords, Sankey; all scales, axes, tooltips, transitions and brushing	VIZ 1–3 + stretch	Wk3, 4, 5, 6, 7
Chart.js	Radar chart (simpler polar layout with built-in legend)	VIZ 5 stretch	Wk 3
Python + pandas	CSV inspection, standardization of ordinal labels, encoding (Never=0...Always=4), export to data.json	Preprocessing	Wk 2
HTML / CSS / JS	Single-page website shell, responsive flexbox layout, global filter panel (dropdowns)	Website skeleton	Wk 1
Observable / Vega-Lite	Rapid exploratory prototyping and sanity checks before D3 implementation	Exploration	Wk 5
Crossfilter.js	Linked brushing: selection on one chart filters all other charts simultaneously	VIZ 3 + stretch	Wk 7

4. Implementation Breakdown

4.1 Minimal Viable Product — required by end of project

Each task below is independent and can be developed in parallel. Together they produce a fully functional visualization page that loads the real Mendeley dataset and answers all three research questions.

#	Task	Expected output	Dependency
1	Data preprocessing	Python script that reads the Mendeley CSV, standardizes all ordinal values, and outputs data.json ready for D3	None
2	Website skeleton	HTML/CSS page with sticky nav, section layout, responsive grid, color system, filter bar UI	None
3	VIZ 1 — Heatmap	D3 heatmap with color scale, tooltips, row/column hover highlight	Tasks 1 & 2
4	VIZ 2 — Grouped bar chart	D3 nested bars, click-to-isolate interaction, legend	Tasks 1 & 2
5	VIZ 3 — Parallel coordinates	D3 multi-axis parallel coords with per-axis brush	Tasks 1 & 2
6	Global filter panel	Gender + year dropdowns wired to re-filter dataset and re-draw all 3 charts	Tasks 3–5

4.32 Current prototype status

A functional website skeleton is already running (see [index.html](#) submitted alongside this report). The page includes the full layout and navigation, a working color system, placeholder containers for all three MVP charts, and a data-loader skeleton. VIZ 1 (heatmap) is implemented with hover interactivity. The bar chart and parallel coordinates scaffolds are in place — they just need the real data.json to be populated. The global filter panel UI is built; filter logic will be connected to the real dataset in the next sprint.

The prototype currently uses simulated data that was deliberately designed to reflect the expected distribution from the Mendeley dataset (e.g. poor sleep quality correlating with lower performance). Replacing it with real data requires only swapping the JSON source — all chart rendering logic is already parameterized.